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Start Date & Time:

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ASEN 5090 Introduction to GNSS

MIDTERM EXAM

October 16, 2025 8AM – October 19, 2025 11PM (MDT)

Format – There are three problems worth a total of 100 points.

You may write directly on the exam sheet and/or use your own paper or tablet computer to show your work. Please number all pages and scan them into a single pdf document for upload to Gradescope. All code used must be included in your submitted pdf.

The name of your pdf file should be: *LastName_FirstName_ASEN5090Midterm.pdf*

Logistics – The exam period will begin on Thursday, October 16, 2025 at 8AM and close on Sunday, October 19, 2025 at 11PM. If you have a question about the exam you may email or text (303-884-1297 include your name) the instructor who will respond promptly to questions between 10:00-11:00AM Thursday, 9:30-10:30AM Friday, 7:00-8:00AM Saturday and Sunday. Once you download the exam from Gradescope, you have up to 24 hours to complete the exam, scan, and upload a pdf of your solutions to Gradescope. However, all solutions must be submitted by the end of the exam period. If you run into difficulty submitting your exam on Gradescope, you may email your solution file to the instructor – but you will still be subject to the 24-hr and end of exam period limits.

Open Book – During the exam, students may use textbooks, class notes, and other references recommended for this course or that you collected/assembled prior to the date of the exam. The only internet site you may access once you begin the exam is our Canvas page. You may use a calculator or computer codes in solving the problems; however, extensive coding or typesetting is not expected.

During the entire exam period (8AM Oct 16 to 11PM on Oct 19) you may NOT work with, consult, collaborate, or discuss the exam with ANYONE except the instructor. Once you have downloaded the exam you may NOT access or search the web, online forums, or resources other than those listed above. Any violation of these exam conditions will result in a grade of F in the course and an honor code violation report will be filed with the honor code office.

If doing calculations by hand, please assume that the speed of light, $c = 3 \times 10^8$ m/s, $\mu_E = 398600.5$ km³/s². If you are using existing code, you may use these values or other established values - make sure to clearly identify the values used in your calculations.

On my honor, as a University of Colorado Boulder student, I have followed all of the above instructions and have neither given nor received unauthorized assistance on this work.

SOLUTIONS

Printed name

PROBLEM 1. (40 points, a-d)

For this problem, **assume GPS orbits are circular with radius $R_{GPS} = 26,560,000$ m and inclination = 55 deg.** The table below gives the WGS-84 ECEF locations of a few satellites in circular orbits (not necessarily GPS satellites) at a particular time.

- a) Compute the magnitude of the velocity for each of the satellites in the table below, assuming they are all in circular orbits. Which satellite would have the largest possible range rate (and largest magnitude Doppler shift) when tracked by observers on the Earth?

- Calculate radius (R_s) of each satellite orbit by finding magnitude of their position vectors

$$v = \sqrt{\frac{\mu}{R_s}}$$

- Sat 2 will have the largest possible range rate – it is a low Earth orbit satellite

- b) An observer (A) is located at N 26 deg, longitude W 80 deg, height above the ellipsoid $h=0$. Compute the WGS-84 ECEF coordinates of A in meters. Express the results (x_A , y_A , z_A) to 1m precision.

$$x_A = 996,102 \quad (\text{m})$$

$$y_A = -5,649,175 \quad (\text{m})$$

$$z_A = 2,779,062 \quad (\text{m})$$

- c) Compute the geometrical range, azimuth, and elevation for each satellite position in the table from observer A's location. Add the appropriate units for these quantities to the table. *(It is not necessary to account for signal time-of-flight in this problem.)* (SEE TABLE)

- d) Which satellites would you expect to be visible to observer A (mark them with a V)?

- Just check for elevations greater than 0.

- Since Sat 2 is so close to 0, it would be equally valid to say that it likely is/is not visible.

Which satellite is closest to the observer's zenith (mark it with a Z)?

- Highest elevation.

Which satellites which could be GPS satellites (mark them with a G)?

Show your work & briefly explain your results – required to get full credit for correct answers.

To be GPS satellites in the specified circular orbits they must have position vector magnitudes of 26,560,000 m. In addition, their orbit planes must have inclination of 55 deg, so the latitude of the satellite (angle of the position vector with respect to the equator) cannot be larger than 55 deg. Sat1 is at latitude of 80 deg, so it cannot be a GPS.

	ECEF X (m)	ECEF Y (m)	ECEF Z (m)	v (m/s)	Range to A (m)	Azimuth (deg)	Elevation (deg)
Sat 1 V (not G)	1,193,698	-4,454,942	26,156,494 incl too high	3,874	23,408,750	1.07	23.20
Sat 2 ~V	832,273	-4,720,052	4,792,867	7,669	2,223,853	0.00	0.90
Sat 3 G	-3,994,192	22,652,188	13,280,000	3,874	30,596,403	0.00	-43.93
Sat 4 V	1,189,069	-36,700,543	0	3,067	37,143,863	140.27	52.37
Sat 5 V, G, Z	7,867,022	-21,614,466	13,280,000	3,874	20,306,890	63.14	77.26

PROBLEM 2. (30 points, parts a-d)

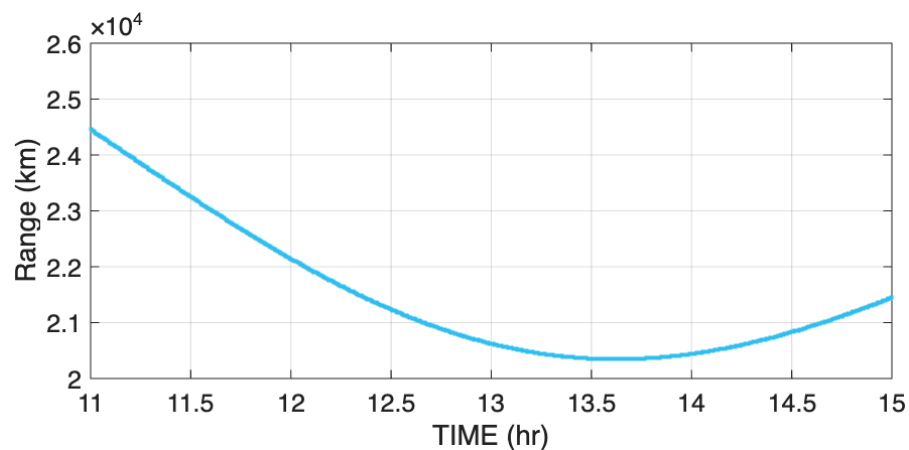
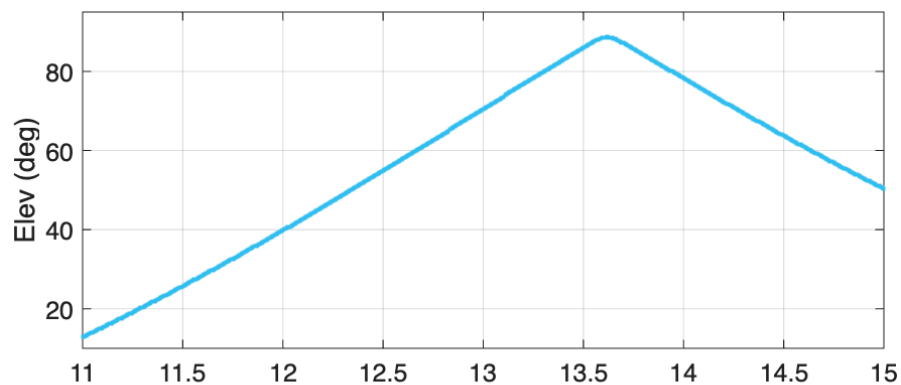
The figures on the following page show the 24-hr skyplot and a 4-hr elevation vs time plot for a location somewhere on the Earth. The elevation vs time plot indicates the specific PRN at its highest elevation. Assume the GPS orbits are circular with radius $R_{GPS} = 26,560$ km and the Earth radius $R_E = 6378$ km.

a) What do you know about the latitude of the location? Explain

- Because the “hole” is at $Az = 180$ deg you know it is in the southern hemisphere
- Because satellites reach elevation of 90 deg, the Latitude must be < 55 deg
- Latitude is closer to the equator than NIST because “hole” is shifting away from overhead compared to NIST (or just try different values with your code from HW2 to see what matches best) – actual latitude is ~ 530 deg

b) Sketch the approximate range from the observer to PRN 27 over this 4-hour period and give approximate max and min values. (Label your axes.)

- Min range = $R_{GPS} - R_E = 20,182$ km
- Max range at elev of 10 deg is 24700 from law of cosines + quadratic formula or the expression given in lecture (and ME) for range as a function of elevation.
Other answers: $24,388$ km for 13 deg, Max range at 0 deg elevation = $25,783$ km

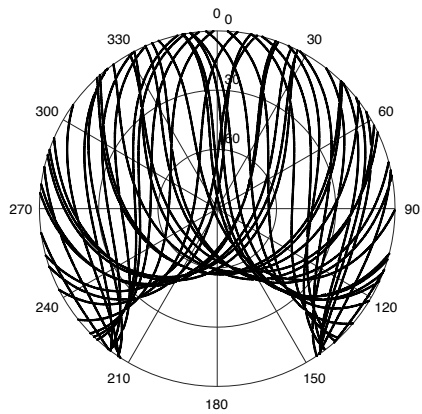


c) At what time within this 4 hour period does PRN 16 have a zero Doppler shift? At what time in this interval does it have its largest positive Doppler shift? Explain how you identified these times.

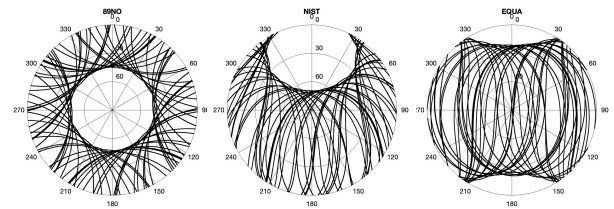
- ~ 12.1 hour, zero Doppler because it reaches its peak elevation and is at point of closest approach.
- At 11 hrs, because as satellite rises, the range is decreasing, typically fastest rate of decrease at the start of the pass => greatest positive Doppler shift.

d) Which satellite would you expect to have the smallest ionospheric delay at t=13:00 hours? Which one is likely to have the largest ionospheric delay? Why?

- PRN27 smallest because it is highest elevation at that time and has shortest path through ionosphere.
- PRN 5 (unlabeled line at the bottom) will have the largest because it has min elevation and longest path through ionospheric layers.
- Also depends on local time of day at ionospheric pierce point, but so could be a slightly higher elevation going through a more active section of the ionosphere



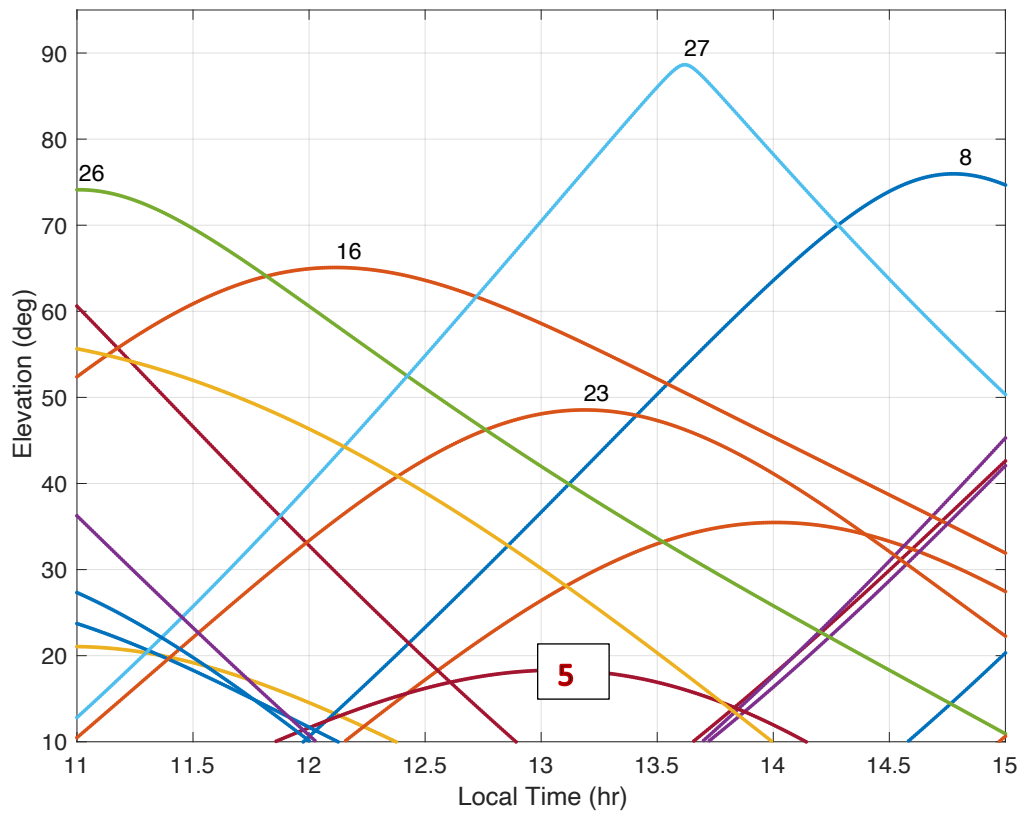
GPS visibility at different latitudes



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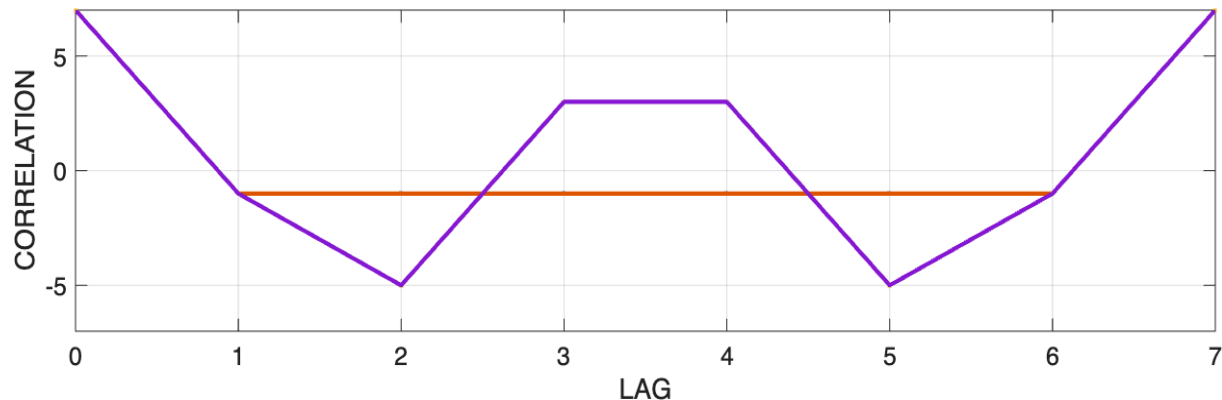
PROBLEM 3. (30 points, a-d)

Two PRN codes, A and B, are considered for a mini ranging system.

$$A = \begin{bmatrix} 1 & 1 & 1 & -1 & -1 & 1 & -1 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & -1 & -1 & 1 & 1 & -1 & 1 \end{bmatrix}$$

The Autocorrelations for Code A & B are shown below:



- a) Compute the auto-correlation for Code B and plot it on the graph above. (in purple)

$$R_{BB} = \begin{matrix} 7 & -1 & -5 & 3 & 3 & -5 & -1 & 7 \end{matrix}$$

- b) What is special about Code A compared to Code B? Is Code A useful for ranging? Explain your answer.

A is a maximal length code, meaning that its value is -1 for all alignments except for 0. (B is not.). Code A is useful for ranging because there is a clear peak at 0 alignment; however, the 7 bit code is very short, so realistically one would not use such a short code.

- c) Would this pair of codes be useful for CDMA? Why or why not?

Useful for CDMA requires low cross-correlation for all offsets: $R_{AB} = -5 \ -1 \ 3 \ -1 \ -1 \ 3 \ 3 \ -5$
 These values are not at all small compared to autocorrelation value of 7, so the codes are not useful for CDMA. (BTW – there is no difference between +/- values because negative correlations would be equivalent to the effect of a data bit inversion.)

- d) If the code chipping rate is 2 MHz:

- What distance (in meters) corresponds to a shift of 1 chip of the A-code?

$$\text{Distance} = c / f_c = 3 \times 10^8 \text{ m/s} / 2 \times 10^6 \text{ chips/s} = 150 \text{ m}$$

- How long (in seconds) does it take to transmit the entire A-code?

$$\text{Period of the code} = 7 \text{ chips} / 2 \times 10^6 \text{ chips/s} = 3.5 \times 10^{-6} \text{ s or } 3.5 \text{ } \mu\text{sec}$$